

Contextual Blind Spots: A Systematic Review of Situational Context in Emotion Recognition Datasets

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Abstract

Emotion recognition systems are beginning to be deployed in critical real-world situations, yet the datasets used to train them remain constrained under controlled conditions with limited attention to the situational factors that play an immense role on how emotions are expressed, perceived, and labeled. This systematic review examines seven recent emotion recognition datasets and their associated studies to survey the extent to which situational context is encoded in current datasets and how modern deep learning architectures can adapt to these specifications. This review identifies three primary gaps in dataset construction which include the absence of contextual data descriptors, demographically classified annotator pools, and reporting standards at dataset release. Finally, a framework for context-aware dataset construction is proposed, with ethical considerations surrounding bias also being considered.

Introduction

Motivation

The ability to recognize and interpret human emotion is a significant component of artificial intelligence due to its ability to enable more nuanced forms of communications and empathetic responses. Computing systems are becoming more deeply embedded in areas where these qualities matter, making the demand for reliable automated emotion recognition grow substantially. Fields including healthcare, education, customer experience, and public safety are all in the process of integrating AI agents into their applications which requires a strong interpretation of emotion to carry out their tasks (Boyd and Andalibi 2023; Pepa et al. 2023). Emotion recognition systems have witnessed a considerable technical advancement over the past two decades progressing from baseline feature extraction methods to deep learning architectures capable of processing facial images, speech signals, EEG data, and text to interpret feelings. By many conventional metrics, the field appears to be maturing and developing steadily.

However, an under examined gap stalls the progress of emotion recognition models as systems that claim to infer emotional states are continuously being deployed. The

conditions under which emotion recognition systems are trained rarely reflect the conditions under which they are utilized in the real-world. In many cases, these deployments have proceeded without any rigorous field-wide evaluation of whether the underlying models are accurate or generalizable. Raji et al. (2022) describes this as an accountability gap, a delay between the usage of automated systems and the emergence of any evaluation infrastructure capable of assessing their actual performance in a real-life context (Raji et al. 2022). Additionally, Capbitza, Campagner, and Mattioli (2022) demonstrate in their paper that the ground truth labels and widely used FER datasets are actually unreliable, reporting an average inter-rater agreement of just $\alpha = 0.44$ which is below the threshold of 0.67 (Cabitza, Campagner, and Mattioli 2022). They present the claim that no architectural improvement can compensate for the constrained ceiling that is imposed by unreliable FER benchmarks. These benchmarks reward accuracy within controlled settings, but in actuality, real-world emotion is not produced in controlled settings. It is shaped by context, who is present, what has been said, and the overarching environment of the situation. This review argues that the field's reliance on de-contextualized training data generates a barrier to reliable emotion recognition, and that the process for resolution requires rethinking not just the model architectures, but the benchmarks that drive them.

The Dataset Problem

Machine learning models are bounded by the data that they are trained on. In the field of emotion recognition, this constraint is particularly important because the majority of the datasets that define the field's benchmarks are poorly suited to real-world deployment.

For a large portion of widely used emotion recognition datasets, their images were constructed under conditions that prioritize experimental control over true validity (Costa et al. 2023). In these datasets, emotional expressions are elicited and exaggerated through scripted scenarios or posed instructions in order to reduce any potential noise and simplify the annotating task. While this may optimize the experimental phase, the datasets produced exclude the variability and ambiguity that characterize emotion as it occurs in reality.

Ultimately, when these trained models are evaluated on naturalistic data, their performance will degrade and produce inaccurate predictions. The emotion detection field's

evaluation practice of training and testing on splits from the same dataset only allow these inaccuracies to prolong and mask the problem rather than address it. Improving the reliability of emotion recognition is fundamentally a data challenge and the solution requires determining whether the target benchmarks are measuring the right features in the first place.

Context Gap

Situational context encompasses several distinct dimensions that will be covered in this review. **Social context** refers to the relational structure of an interaction, capturing who is speaking to whom and the nature of their relationship. **Environmental context** refers to the physical setting and domain of the interaction. **Cultural context** reflects the display rules or norms that govern emotional expression within a given situation and recognition within a given cultural community. **Temporal and sequential context** covers what has been said or experienced prior to the current moment that helps frame the present emotion. Together, these contextual features constitute the situational surrounding that provides meaning to one's emotion.

Current datasets either encode these features poorly or not at all. Models are left to infer emotional meaning from signals stripped of the situation and results in systems that struggle to interpret naturalistic emotion. This review aims to focus on the role of these individual contexts in dataset design, model reliability and performance to provide advice for future datasets to generate more accurate emotion detection performances in real-world settings.

Research Questions

This systematic review addresses the following research questions:

1. To what extent do current emotion recognition datasets encode situational context, including social, cultural, environmental, and socio-demographic factors?
2. Which model architectures are best suited to leverage context-rich training data?
3. What methodological gaps in current dataset construction practices hinder the development of context-aware and demographically representative emotion recognition benchmarks?

Related Works

Previous systematic reviews of emotion recognition have already conducted analysis on potential performance improvement techniques that can be employed on deep learning systems. While they provide useful insights, the relationship between situational context, dataset design, and model reliability is left underexplored.

Pereira et al. (2024) presented an SLR for the emotion recognition field to shed light on computer vision and deep learning methods, mainly highlighting the advantages of using ViTs, CNNs, and faster R-CNNs (Pereira et al. 2024). While this review provides a rigorous architectural landscape, its organizational framework does not evaluate

datasets against the contextual metadata they consume. The introduction even explicitly states the challenges that arise from expression recognition in real-world scenarios, yet the paper continues to treat the test environment as a binary distinction between controlled and uncontrolled conditions.

Pereira, Moniz, and Carvalho (2024) provide a comprehensive survey of deep emotion recognition in textual conversations to document how transformer-based language models and GNNs have become more dominant for modeling connections between utterances. This survey delves deeper into the importance of context as it identifies conversational context as an essential modeling feature. Conversational context relates to the retention of prior and surrounding utterances to inform emotion classifications. The author's ultimately found that transformer architectures outperformed recurrent networks in capturing long-range sequential dependencies and found them to be better models for interpreting conversational settings (Pereira, Moniz, and Carvalho 2024). This work relates more to this paper's target of the architecture-context relationship, yet is limited to the temporal and sequential structure of dialogue. Missing dimensions like social setting, cultural norms, and demographics are all still significant to a model's performance with its corresponding dataset.

In regards to the importance of demographic representation, Verhoef and Fosch-Villaronga offer a direct engagement with the bias dimensions of emotion detection benchmarks. Their research analyzes a broad corpus of affective computing datasets and documents and they showcase substantial underrepresentation across race, sex, and gender, warning that systems trained on widely used datasets are likely to exhibit corresponding biases when deployed in real-world systems (Verhoef and Fosch-Villaronga 2023). They further call for more standardized demographic documentation and inclusive sampling strategies which aligns with our key motivation that emotion detection methods with low performance stem from generally unreliable datasets.

Methodology

Review Protocol

This review was conducted along the guidelines set by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) (Garrison, Nweashe, and Qu 2026; Konkar and Qu 2025; Xie et al. 2026; Murungi et al. 2023).

Inclusion Criteria A paper was included if it satisfied all of the following conditions: (1) the study addressed emotion recognition, affect recognition, or closely related tasks (FER, speech emotion recognition, or conversational emotion recognition) as a primary focus; (2) the study presented an original emotion recognition dataset, evaluates a model on one or more existing benchmarks, or analyzes the properties of existing datasets; (3) the study employed a machine learning or deep learning method; (4) the study addressed either situational context in their dataset design, demographic properties of subjects, or bias in emotion recognition systems; (5) The date range specified for the paper selection process was between the years 2015 and 2025.

Exclusion Criteria A paper was excluded if any of the papers met the following conditions: (1) it focused exclusively on sentiment analysis without addressing emotion recognition; (2) the paper was purely conceptual without any empirical data; (3) the paper discussed emotion from a purely psychological perspective without a computational contribution.

Search Strategy

Two main databases were utilized for systematic searching: the ACM Digital Library and Semantic Scholar. The ACM Library was selected for its ability to incorporate Boolean operator within the search bar and its overall comprehensive coverage of computer science research topics. Semantic Scholar was selected as a broader database within the CS field to capture any publications that were barely filtered out by the ACM search field.

The following Boolean string was used for the search process:

("Emotion Recognition" OR "Affect Recognition" OR "Speech Emotion Recognition" OR "Facial Expression Recognition" OR "Conversational Emotion Recognition") AND ("Dataset" OR "Benchmark" OR "Corpus") AND ("Situational Context" OR "Social Context" OR "Contextual Information" OR "Cultural Context" OR "In-the-wild" OR "Racial Bias" OR "Gender Bias" OR "Annotator Bias" OR "Demographic Diversity" OR "Display Rules") AND ("Context-aware" OR "Graph Neural Network" OR "Self-attention" OR "Multi-modal Fusion" OR "Dialogue model" OR "Transformer" OR "Attention Mechanism")

The string is organized into four thematic blocks. The first block targets the core topic domain of emotion recognition. The second block restricts results to studies with a dataset or benchmark component. The third targets contextual and demographic dimensions which are essential to locating the datasets that are critical for our research. The fourth restricts the architecture vocabulary to models with explicit contextual modeling capacity.

Screening Process

Before screening each paper, a merged citation file with all sources was imported into JabRef to detect any duplicates between both databases. Once those papers were eliminated, Covidence was used to screen the papers individually in three stages:

(1) In the first stage, all retrieved records were assessed at the title level, in addition to inspection of author-assigned keywords where titles were ambiguous or unclear. Papers were excluded at this stage if the title and keywords both indicated that the study did not address the inclusion criteria, or showcased that they met an exclusion criteria. If any uncertainty existed about whether a page was relevant, it was passed to the second screening stage rather than excluded.

(2) For the second stage of screening, papers were then assessed at the abstract level against the full inclusion and

exclusion criteria. Particular attention was given to distinguishing emotion recognition from sentiment analysis, verifying the existence of a dataset, and confirming the usage of a computational method.

(3) The final stage involved obtaining the full texts of the remaining papers and analyzing each of them with the same inclusion and exclusion criteria. The most time was spent with this stage to ensure that every paper met the specific information constraints of the requirements set in place.

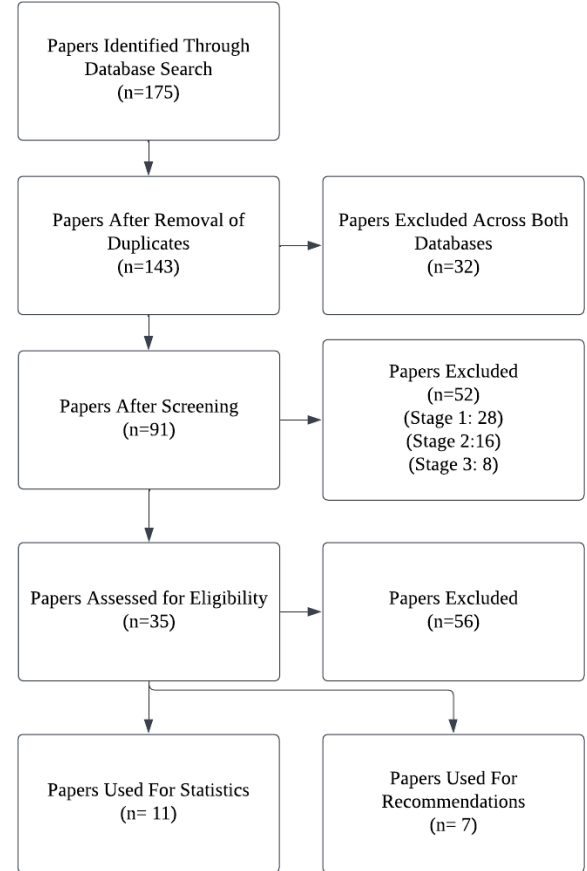


Figure 1: PRISMA flow diagram for study selection.

Relevant ML Models

CNN Convolutional Neural Networks (CNN) are a class of important machine learning models that are particularly apt for handling image-based classification. This is due to the model’s ability to learn spatial features from a pixelated grid at high speeds (Jaiswal and Nandi 2020). A CNN processes images through a series of convolutional layers that each apply a set of filters made to detect patterns within the image. This moving filter makes them particularly advantageous for image data as it allows for the detection of nuanced patterns like the curvature of a smile or the raise of an eyebrow. The drawback for CNNs in the scope of this research is that their contextual capacity is inherently limited

Type	Criterion	Include?
Inclusion	Focuses on emotion recognition, affect recognition, or related tasks (FER, SER, conversational emotion recognition)	Yes
	Uses or introduces an emotion recognition dataset, evaluates models, or analyzes datasets	Yes
	Employs machine learning or deep learning methods	Yes
	Addresses context, demographics, or bias in emotion recognition systems	Yes
	Published between 2014 and 2025	Yes
Exclusion	Focuses exclusively on sentiment analysis without emotion recognition	No
	Purely conceptual (no empirical data or experiments)	No
	Purely psychological discussion without computational contribution	No

Table 1: Inclusion and exclusion criteria for paper selection in the systematic review.

by receptive field size. A CNN model processes emotion as a function of localized features rather than situational narrative, meaning it can recognize a frown but struggles to interpret why that frown is occurring. Nonetheless, CNNs are a strong baseline for feature extraction in a hybrid emotion detection model (Zheng, Mendieta, and Chen 2023) but a weak standalone architecture for benchmarks where situational context is a distinct factor.

Transformers The transformer architecture has become a significant foundation for recent LLM models and ML systems that process both sequential and spatial data. In the emotion detection field, vision transformers (ViTs) and BERT have proven to be relevant in their own respective domains where the ViT models has proven to be proficient with image-based inputs and BERT with text-based data. They are currently the most context-aware models through their utilization of self-attention or cross-attention. This feature allows them to model long-range dependencies across an entire sequence simultaneously, enabling the model to weigh the emotional significance of one utterance against all others in a conversation or scene (Ezzameli and Mahersia 2026). Models like BERT, RoBERTa, and their multi-modal variants have all demonstrated that pretraining on large contextual benchmarks substantially improves performance as transformers have proven to be “data hungry” architectures (Liu et al. 2019). The main limitation of transformers is that attention is bounded by token context windows and can be computationally expensive over long sequences.

RNN/LSTM Recurrent neural networks are a model that process sequential data by retaining an internal memory that updates at each time step. RNNs use both current inputs and past hidden states to make predictions, providing them a built-in mechanism for sequential situational context. Earlier events in a conversation or scene may inform the interpretation of later emotional states, so holding that data in memory is a major advantage (Majumder et al. 2019; Shou et al. 2026). A key drawback of RNNs is that they suffer from vanishing gradients over long sequences, meaning distant context is forgotten. Long short-term memory (LSTM) networks are a specialized type of RNN that use a unique internal memory cell to overcome the vanishing gradient problem. The flow of information is regulated by the forget, input, and output gates each with their own ac-

cess control over what information should be stored and what should be discarded. Retaining data for longer periods of time only accumulates the amount of context that can be stored, which is beneficial to processing datasets that are abundant in situational information.

Results

RQ1: Dataset Landscape

The screening process yielded eight studies, each presenting an original emotion recognition dataset contribution with explicit engagement to real-world, contextually sound data. Table 1 provides a high-level overview of the included corpus. Publications span from 2020 to 2026, with six of the seven studies published between 2023 and 2026. Two studies (Jang et al., 2026; Deschamps-Berger et al., 2023) do not primarily contribute a new dataset but use existing datasets as the empirical basis for investigating contextual and architectural questions, so they are included for their direct relevance to the review’s research questions. Table 1 provides a structured summary of all the included studies, their datasets, publication venues, modalities, elicitation methods, and relevant domains.

Modality All seven studies engage with multi-modal data to varying degrees, most often combining text, audio, and video together. This combination appears in MERR (Cheng et al. 2024), HCI-ERC (Kovacevic et al. 2024), and Amd’SaEr (Haouhat et al. 2025). K-EmoCon (Park et al. 2020) extends this further by adding EEG signals, delving into physiological data of humans making the dataset more rich in context. CEMO (Deschamps-Berger, Lamel, and Devillers 2023) is the most constrained dataset, relying on audio and manual text transcriptions only, reflecting how real-world scenarios like telephone-based emergency calls are limited within their field. FindingEmo (Mertens et al. 2024) and the cross-cultural study by Jang et al. operate exclusively with visual image data, though they do differ in scale.

Elicitation Method Three broad elicitation categories are showcased across the included studies: naturalistic collection, media data, and the re-annotation of existing datasets. The majority of the studies employed naturalistic or semi-naturalistic collection. FindingEmo scraped images from

Study	Year	Venue	Dataset	Modalities	Size	Elicitation	Domain
Park et al.	2020	Sci. Data	K-EmoCon	AV + EEG + Physio + Text	32 participants	Naturalistic	Social conversation
Deschamps-Berger et al.	2023	ICMI	CEMO	Audio + Text	20+ hrs; 4,224 segments	Naturalistic	Emergency call center
Cheng et al.	2024	NeurIPS	MERR	Video + Audio + Text	33,105 samples	Acted (TV/film)	Media
Kovacevic et al.	2024	ICMI	HCI-ERC	Video + Audio + Text	99 participants; 350+ hrs	Naturalistic	Human–chatbot HCI
Mertens et al.	2024	NeurIPS	FindingEmo	Images (visual)	25,869 images; 655 annotators	Naturalistic (web)	In-the-wild social scenes
Haouhat et al.	2025	ACM TAL-LIP	Amd’SaEr	Video + Audio + Text	1,037 samples	Naturalistic	Social media
Jang et al.	2026	IUI	Cross-cultural	Images (visual)	70 participants; 200 images	AffectNet (re-annotated)	Cross-cultural annotation

Table 2: Summary of reviewed emotion recognition studies.

the open web using emotionally-themed search queries, K-EmoCon recorded live debates between randomly paired participants, CEMO captured authentic emergency calls to a French call center, HCI-ERC recorded participants interacting with a real GPT-based chatbot, and Amd’SaEr collected Arabic social media videos. MERR took clips from movies and TV series that were then annotated by performers while Jang et al. re-annotated images from AffectNet, a pre-existing facial expression benchmark.

It is important to note that there is a significant amount of variation within ecological validity of the data collection process. For example, the HCI-ERC data was collected over three weeks in participants’ own environments, whereas K-EmoCon debates took place in a controlled university room.

Application Domain The included studies span five specific application domains. FindingEmo carries the broadest domain which are images depicting various social scenes drawn from the internet. CEMO, on the other hand, is narrowly specified as it represents one of very few publicly available datasets targeting emergency response contexts where emotion serves as a function of safety. K-EmoCon and HCI-ERC both operate within conversational contexts with one representing human-human debates while the other analyzes human-AI chatbot interactions. MERR and Amd’SaEr draw on entertainment and social media sources. The Jang et al. cross-cultural study examines annotation practices rather than operating within a new application. Papers targeting domains like learning environments or customer service contexts were expected but their absence may represent a gap in the reviewed literature.

Contextual Coverage

Mapping Datasets Against Context Taxonomy Relating back to the four-dimensional context taxonomy (social, environmental, cultural, and temporal/sequential con-

text) mentioned in the introduction, each dataset has been mapped with these categories, along with a fifth section of socio-demographic reporting. The full comparative research demonstrated that no single dataset addressed all four contextual dimensions all together. Temporal/sequential and social context received the most attention, while cultural context remained almost universally unaddressed.

Social Context Social context is the most commonly represented dimension across the included papers as it was present in some form in all of the seven datasets. FindingEmo (Mertens et al. 2024) was the only dataset that targeted whole-scene social interactions as a primary annotation goal. The images that were included depicted multiple people in naturalistic social settings and rather annotated the emotional content of the entire scene rather than any single individual. In the paper, the authors illustrate why this design choice was necessary with an example of a bride at a wedding whose facial expression in isolation appears distressed. However, when the smiling faces of surrounding participants and the overall setting are included, it is clear that she is actually overwhelmed with emotion. This positions FindingEmo as the sole dataset in the review that takes advantage of scene-level predictions rather than individual-level perception.

K-EmoCon (Park et al. 2020) advances social context in a slightly different approach by capturing emotion from all three available perspectives within a dyadic (interaction between two individuals) interaction. The first perspective is the experiencing subject, the second is the interaction partner, and the third would be the external observers. This three-dimensional design is unique among included studies and directly encodes the relational aspects of emotion as a social occurrence.

CEMO (Deschamps-Berger, Lamel, and Devillers 2023) and HCI-ERC (Kovacevic et al. 2024) both include social

context through bi-directional interactions between different roles. CEMO captures the emotional dynamic between callers in crisis and emergency operators, noting that callers and agents exhibit distinct emotional profiles. Callers will exhibit feelings of fear, anger, and sadness, while agents will maintain neutral or positive states. HCI-ERC introduces another interaction, not between humans, but with humans and chat-bots which delves into a more artificial emotional relation. The finding that emotions measured from the human’s self-report were more difficult to predict than the human’s perception of the chat-bot’s emotion aligns with K-EmoCon’s observation about the predictability of externally perceived and internally experienced emotions.

Environmental Context Environmental context is addressed most heavily in MERR and CEMO. In MERR, environmental context is an explicitly designed component of the annotation process. Peak emotional frames are analyzed and descriptions of activities and setting are extracted from them to generate a Visual Objective Description that feeds into the final multi-modal annotation. This makes MERR the only dataset in the review where environmental context is formalized as a labeled annotation dimension. For CEMO, the emergency call center environment is itself a richly contextualized area, and the diversity of caller types and people documented reflects environmental variation that other datasets lack.

Cultural Context Cultural context is the most consistently absent dimension across the reviewed datasets, with one exception. Jang et al. (2026) is the only study where cultural context is the primary research question, and what they discovered after performing a re-annotation was that existing large-scale FER benchmarks like AffectNet actually exhibit significant cultural bias. South Korean participants labeled significantly more images as surprise than American participants, indicating that the same facial configurations carry different emotional valence interpretations across cultures. Their study further reveals that Koreans predominantly frame ‘surprise’ situations as negative, while Americans report a more balanced distribution. This showcases that cultural context is not just a demographic variable but a systematic indication of annotation validity.

Temporal/Sequential Context K-EmoCon provides the richest temporal structure since emotions are annotated at five-second intervals through each ten-minute debate. During these windows, physiological signals are captured at high sampling rates across a multitude of modalities. MERR encodes temporal context within clips through the VideoMAE temporal encoder and is able to capture dynamic facial expression sequences through its peak-frame selection procedure. The HCI-ERC dataset partially accounts for temporal context as user-specific data was accumulated over ten sessions that were separated within a three week period.

Socio-demographic Context Socio-demographic reporting across the included studies were inconsistent and often incomplete. K-EmoCon provided the most demographic documentation, reporting the age, gender, and nationality for

all 32 of their participants. They also reported the same information for the five external raters. CEMO’s dataset involved 870 distinct callers and 7 agents, but caller demographic data was not reported or analyzed, despite its importance to understanding which populations are represented in emergency call data. In the end, no study tested whether annotators of different ages, genders, or cultural backgrounds labeled emotions differently within the same dataset.

RQ2: Impact on Model Reliability

Architectural Overview Table 3 summarizes the model architectures employed across the included studies and the mechanisms by which contextual information is incorporated into them. Six of the seven studies deployed or evaluated deep learning architectures, with transformer-based models dominant across text (RoBERTa, FlauBERT, LLaMA2, AraBERT), audio (HuBERT, DistilHuBERT, wav2vec2) and video (EfficientNet, MAE, VideoMAE, EVA, DINOv2) modalities (Hsu et al. 2021; Chang, wen Yang, and yi Lee 2022; Baevski et al. 2020; Tan and Le 2020).

After reviewing the individual architectures, there were two model designs that emerged as most effective for leveraging contextual information. The strongest performance gains that could be attributed to contextual modeling was demonstrated by the Emotion-LLaMA architecture (Cheng et al. 2024). The model was pretrained on 28,618 annotated samples before fine-tuning on 4,487 expert-annotated descriptions. The multi-view visual encoding strategy which captures local facial expression (MAE), temporal facial dynamics (VideoMAE), and global background context (EVA) all in one, significantly outperforms any single-view method.

For sequential and social context, the Bi-GRU architecture in HCI-ERC (Kovacevic et al. 2024) demonstrated the value of having parallel Bi-GRU networks for the user speaker, chat-bot speaker, and the global conversation as it outperformed single-GRU baselines consistently. The key finding in the CEMO architecture (Deschamps-Berger, Lamel, and Devillers 2023) was that same-speaker conversational context (71.7% UA) outperformed cross-speaker context (70.2% UA), suggesting that self-consistency in emotional state is a stronger predictive signal than the response of the other speaker.

A recurring finding across multiple studies was the under performance of video-based modalities compared to text and audio, which contradicted the notion that video carried the richest contextual information. HCI-ERC demonstrated this most clearly, showing that when they added video to text-only models it only added marginal gains (+3.12 macro F1) and sometimes even diminished performance. This could be associated with reduced facial expressivity that was found in human-chat-bot interactions compared to human-human acted datasets like IEMOCAP. This finding showcases that the contextual value of a modality is domain dependent.

Study	Architecture	Context Modeling	Best Result
Mertens et al.	VGG16, ResNet, DenseNet, CLIP, DINOv2	Late fusion of CNN/ViT features	VGG16, DINOv2
Cheng et al.	LLaMA2-7B + LoRA, HuBERT, MAE, VideoMAE, EVA	Instruction tuning, coarse-to-fine pretraining on MERR	F1=0.9036 (MER2023)
Deschamps-Berger et al.	FlauBERT, wav2vec2-FR-3k, Transformer encoders	Multi-scale context window	UA = 71.7% (text + same-speaker context)
Kovacevic et al.	RoBERTa + DistilHuBERT + EfficientNet	Separate Bi-GRUS for user/chatbot speaker states	VAD macro-F1 41.72% (text+audio)
Haouhat et al.	MLP baseline, Graph_MFN	Graph-based multi-modal fusion (Graph_MFN), attention mechanisms	F1_Emo = 0.614, Acc_sent = 73.9%

Table 3: Key model architectures used across included studies

Discussion

RQ3: A Proposed Framework

The results of the contextual coverage reveals that no single dataset in the reviewed literature addresses all four contextual categories simultaneously, so drawing on the strengths and limitations of the reviewed studies, we propose a framework for context-aware emotion recognition dataset construction.

First, contextual metadata should be captured at collection time rather than inferred. The MERR data collection process demonstrates that both environmental and contextual descriptors can be extracted automatically (using MiniGPT-v2 for scene context) and then shifted towards labeled annotation dimensions. FindingEmo’s annotators rate Valence, Arousal, and Emotion but not the specific contextual features that inform their emotional judgment. Social role of participants, setting, task, and perceived dynamic between speakers all play a role in contributing to beneficial data annotations.

Second, subject demographic stratification should be a design requirement rather than a post-collection statistic. The cross-cultural study by Jang et al. (2026) demonstrates that recognition and valence interpretation of surprise differs between South Korean and American annotators, with effect sizes large enough to weaken the validity of majority-vote labels in cross-cultural use cases. A standard for demographically representative dataset construction includes specified sampling for age, gender, ethnicity, and cultural/linguistic background, for subjects appearing in the data and for the annotators labeling it (Lukac et al. 2023).

Third, annotator pool diversity should be emphasized with the same status as inter-rater reliability (IRR) reporting (Dominguez-Catena, Paternain, and Galar 2024). The reviewed papers employ either majority-vote or mean-aggregation across annotators without testing whether annotator background is a source of variance. This approach can only be seen in Jang et al. where they retained annotation distributions rather than oversimplifying them into a single label. Soft label distributions that preserves annotator disagreements provides substantial information and are better suited to training models on culturally contingent emotional judgments.

Fourth, reporting standards for datasets should include a minimum demographic profile (age, gender, cultural background, and professional context). An explicit statement of which contextual dimensions are captured, which are absent, and which are potential sources of bias should also be included in addition to the benchmark. K-EmoCon came the closest to inclusion, with its heatmaps disaggregated by participant and perspective, however, none of the reviewed papers provided all of these variables at release.

Ethical Considerations

The reviewed studies raises ethical concerns that warrant explicit attention in the process of dataset construction and deployment. The most critical ethical concern is the affect on marginalized groups when biased emotion recognition systems are deployed in real-world settings. Emotion recognition is increasingly proposed for use in hiring screenings, mental health monitoring, and educational assessment, domains where misclassifications of emotional state for particular demographic groups can cause harm (Hosseini, Fard, and Mahoor 2025; Fan, Xiao, and Washington 2023).

Conclusion

This systematic review analyzed whether current emotion recognition benchmarks are adequately designed to include situation context that ultimately shapes emotional expressions in real life. Through the examination of seven representative studies, it has been found that the existing gaps between performance and datasets are not incidental but structural, reflecting assumptions about how emotion can be reliably measured. Studies that prioritize naturalistic data collection (FindingEmo, HCI-ERC, CEMO) achieve higher real-world accuracy but face challenges with annotation reliability and demographic representation. On the other hand, studies that prioritize control (K-EmoCon, MERR) gain technical precision at the cost of naturalness. The proposed framework is a practical response to the lack of contextual metadata at collection, demographic classification of subjects and annotators, and proper documentation of data collection methods. These standards are achievable for constructing reliable datasets and will be integral for the next wave of context-aware emotion recognition models.

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